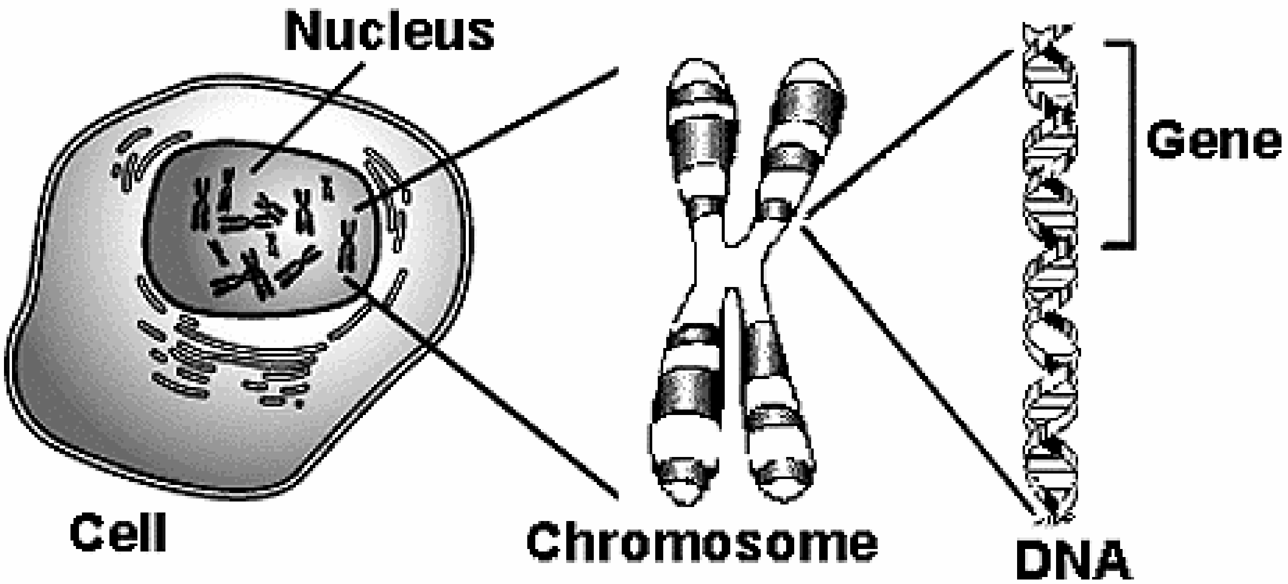
Important Directions

- Read through all slides
- Watch all videos

Genetics and Prenatal Development



A **gene** is a segment of DNA that codes for a particular protein. These proteins determine what we are like. Different versions of the same gene are referred to as **alleles** (e.g., black hair vs. brown hair).

Genes

- Almost all human characteristics are a result of interaction between numerous genes.
- Most inherited human traits are determined by more than one pair of genes.
- One allele (form of the gene) is inherited from mother and one from father.
- The dominant gene is expressed while the recessive gene is not expressed.

Genotype vs. Phenotype

- Genotype: All the genetic possibilities an individual inherits
- Phenotype: The trait that is actually expressed in an individual

Dominant	**Recessive**
Curly hair	Straight hair
Dark hair	Blonde hair
Facial dimples	No dimples
Normal hearing	Deafness (some forms)
Normal vision	Nearsighted vision
Freckles	No freckles
Unattached ear lobe	Attached ear lobe
Can roll tongue in U-shape	Cannot roll tongue in U-shape

Ashley

Ashley's Parents

Mom's genetic contribution:
brown hair, light eyes, IQ above average, behavioral inhibition

Dad's genetic contribution:
blonde hair, light eyes, IQ below average, behavioral inhibition

Ashley's Genotype

Genotype: The genes a person has for a trait.

Brown vs. Blonde Hair

Light eyes

Below Average vs. Above Average IQ

Behavioral Inhibition

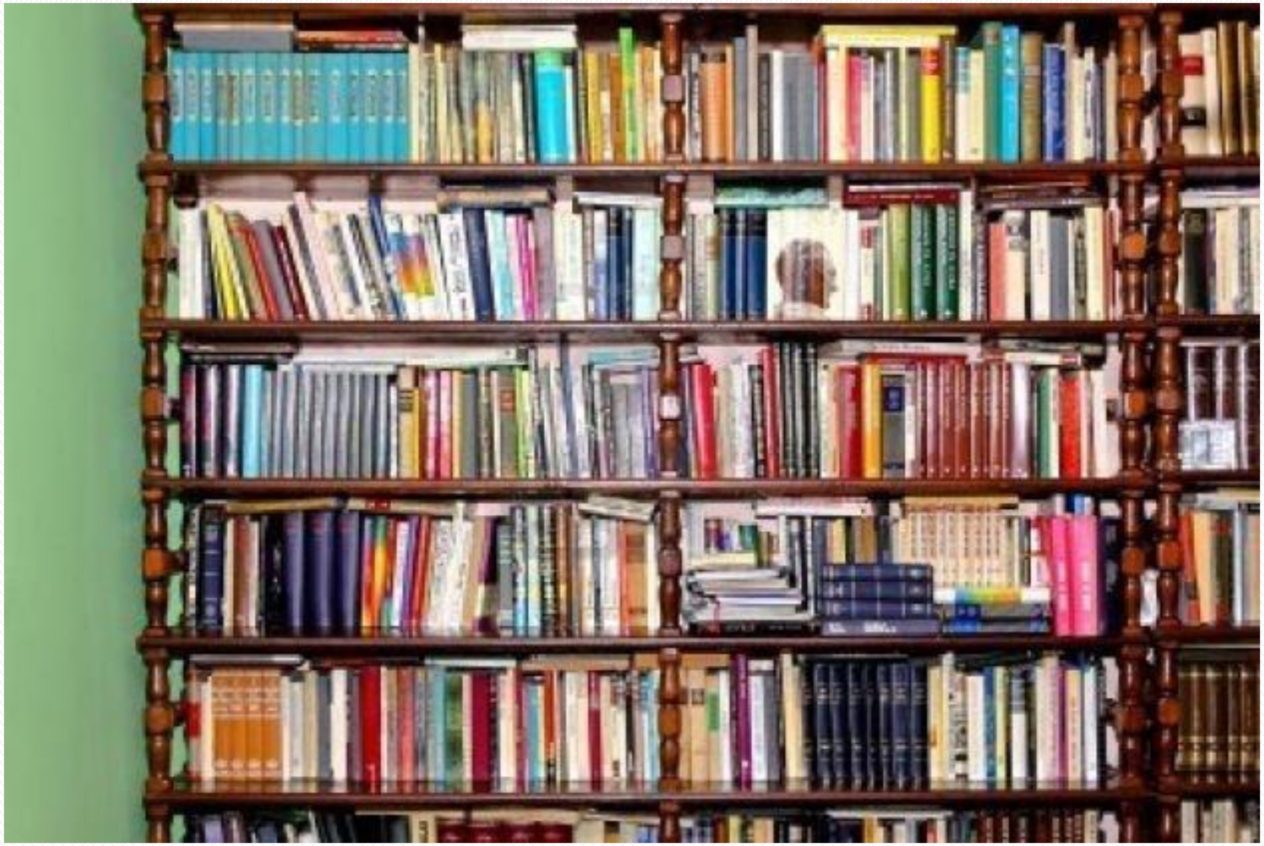
Ashley's Phenotype

Phenotype: What will be expressed.

Ashley has brown hair, light eyes, an above-average IQ, and is behaviorally inhibited

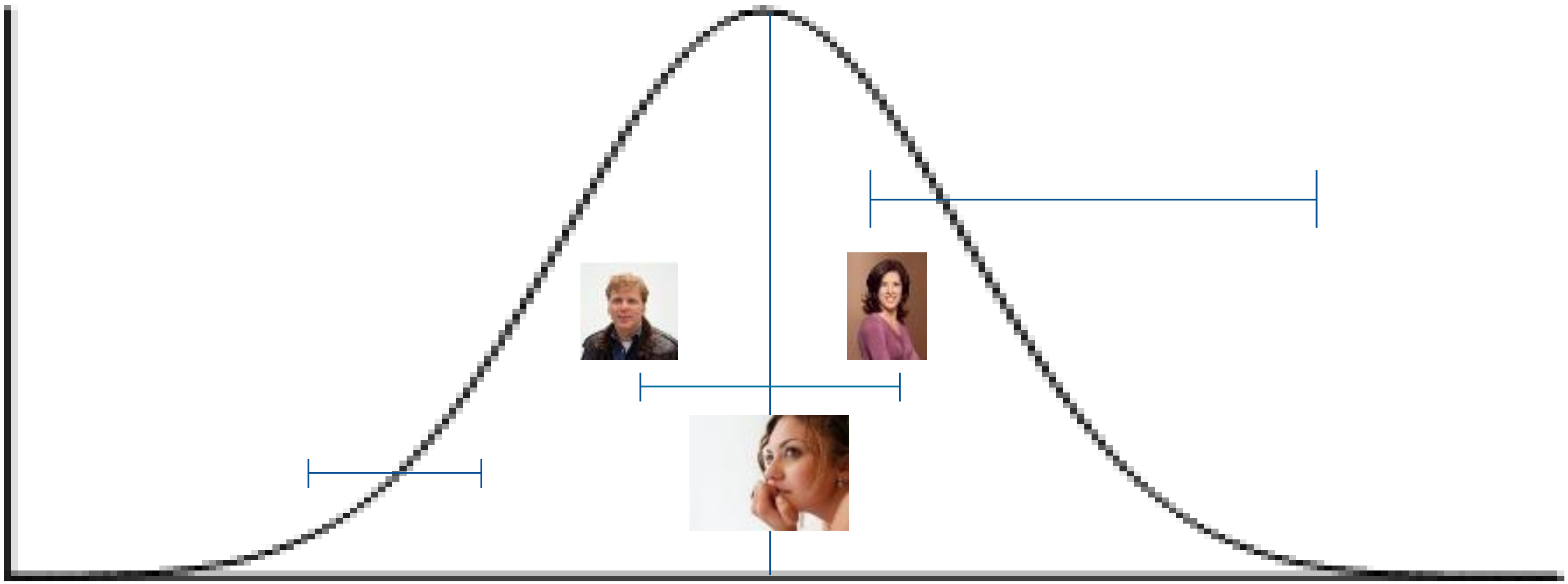
How did Ashley end up with an above-average IQ?

Where Ashley Grew Up



Reaction Range

The variety of phenotypes possible given a particular genotype.



- The genotype sets up the high and low boundaries for the possibilities for the phenotype.
- The environment determines where you fall within that range.
- You inherit a range of possibilities, but the environment determines what your phenotype will be like within that range.

Epigenesis and Reaction Ranges

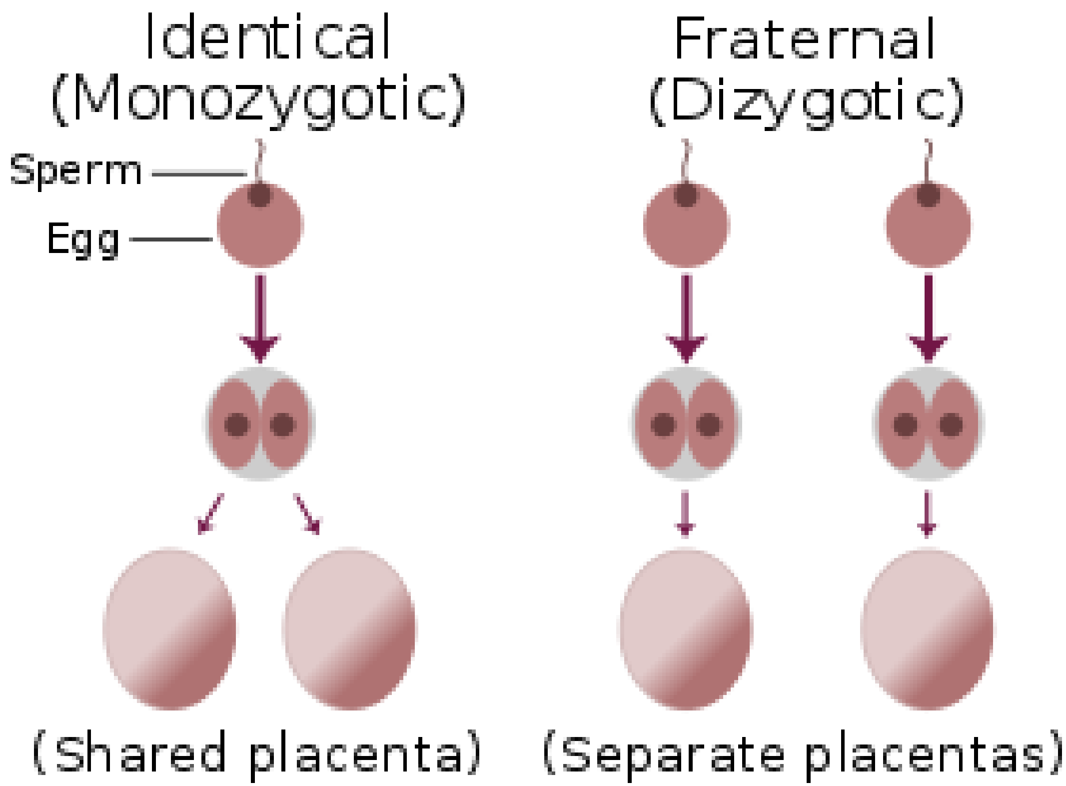
DNA is not destiny: WATCH DOCUMENTARY IN THE PARTICIPATION FOLDER

- Epigenesis – the continuous bidirectional interactions between genes and environment
- Genetic activity responds constantly to environmental influences
- Genes establish a reaction range of potential expression and environment determines where phenotype will fall

Behavioral Genetics

- Attempts to understand “how much” of a trait is due to genes or environment
- Compares relatives sharing different degrees of genes to see if they share a trait
- Identical twins>DZ twins>Cousins> Strangers

Monozygotic and Dizygotic Twins



Identical twins>DZ twins>Cousins> Strangers

Twin Studies:
Concordance Rates

Emily and Steph are identical (MZ) twins. If Emily develops schizophrenia, there is a 41-65% chance that her sister, Steph will also develop schizophrenia.

BUT

Twin Studies:
Concordance Rates

Mary and Steve are fraternal (DZ) twins. If Mary develops schizophrenia, there is a 0-28% chance that her brother, Steve will also develop schizophrenia.

Heritability

- Extent to which differences in any trait within a population are attributed to inheritance
- Value from 0-1
- Intelligence 0.52
- 52% differences between people's IQ in a given population is due to genetics.
- Says NOTHING about one person. Says something about average diff between people in a sample.

- The rest of the 48% of individual differences in IQ comes from ________________
- Environments
- There are also Gene X Environments
- Importance of genes varies with environment

Heritability of IQ varies with Resources in Environment

- When children's environment is rich, IQ is more dependent on differences in genes.
- Childhood environment is supportive and rich.
- Therefore, differences in environment within this population matters lesser.

Heritability of IQ varies with Resources in Environment

- When environment is less supportive, IQ is more dependent on environmental differences.
- Much more variability in environments experienced.
- Individual differences in experience will exert greater influence

How genes influence environments

- 1) Passive effects of genes on environment
- Results from the fact that in a biological family, parents provide both genes and environment to their children
- Decrease with age

- 2) Evocative effects of genes on environment
- Results when a person's inherited characteristics evoke responses from others in the environment
- Stay same across age

- 3) Active effects of genes on environment
- Increase with age
- Results when people seek out environments that correspond to their genotypic characteristics

Conception

- For conception to occur there must be a released ovum and a sperm
- Ovulation releases the ovum, and if sperm is available, fertilization can occur

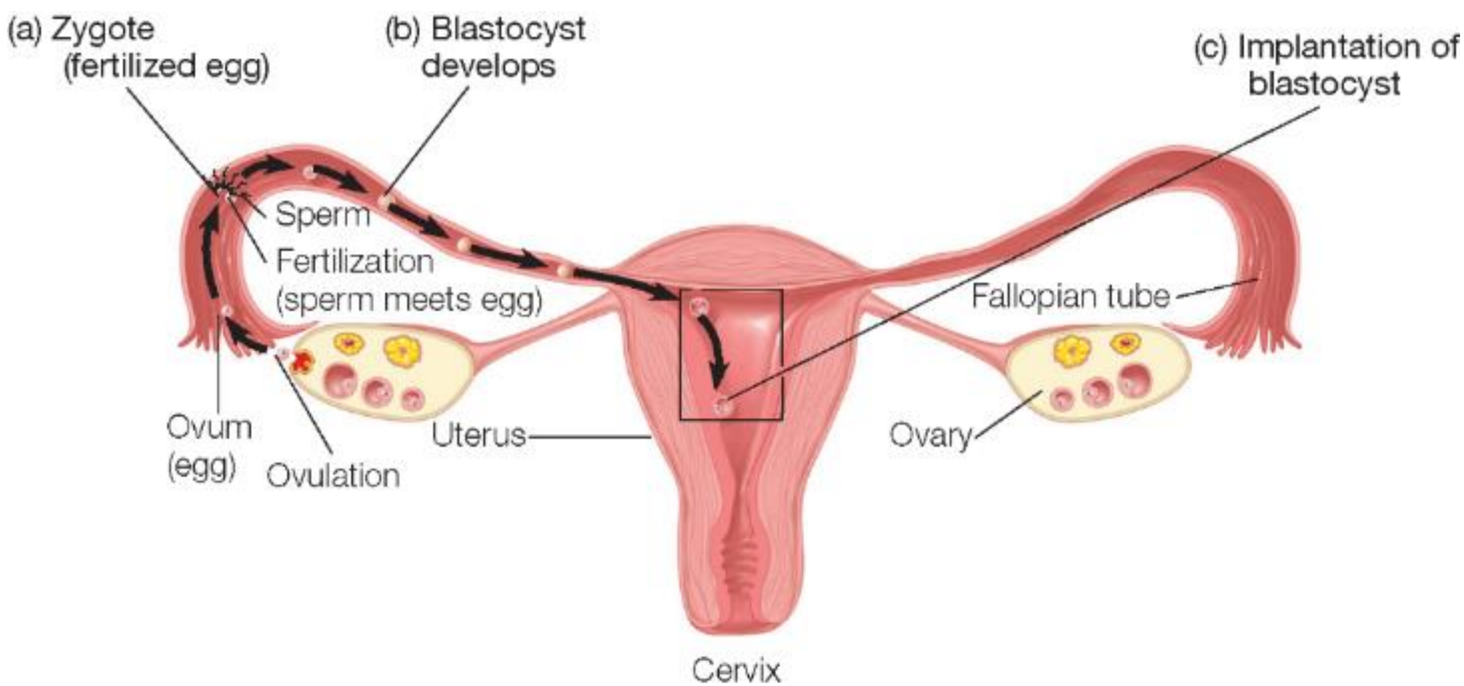
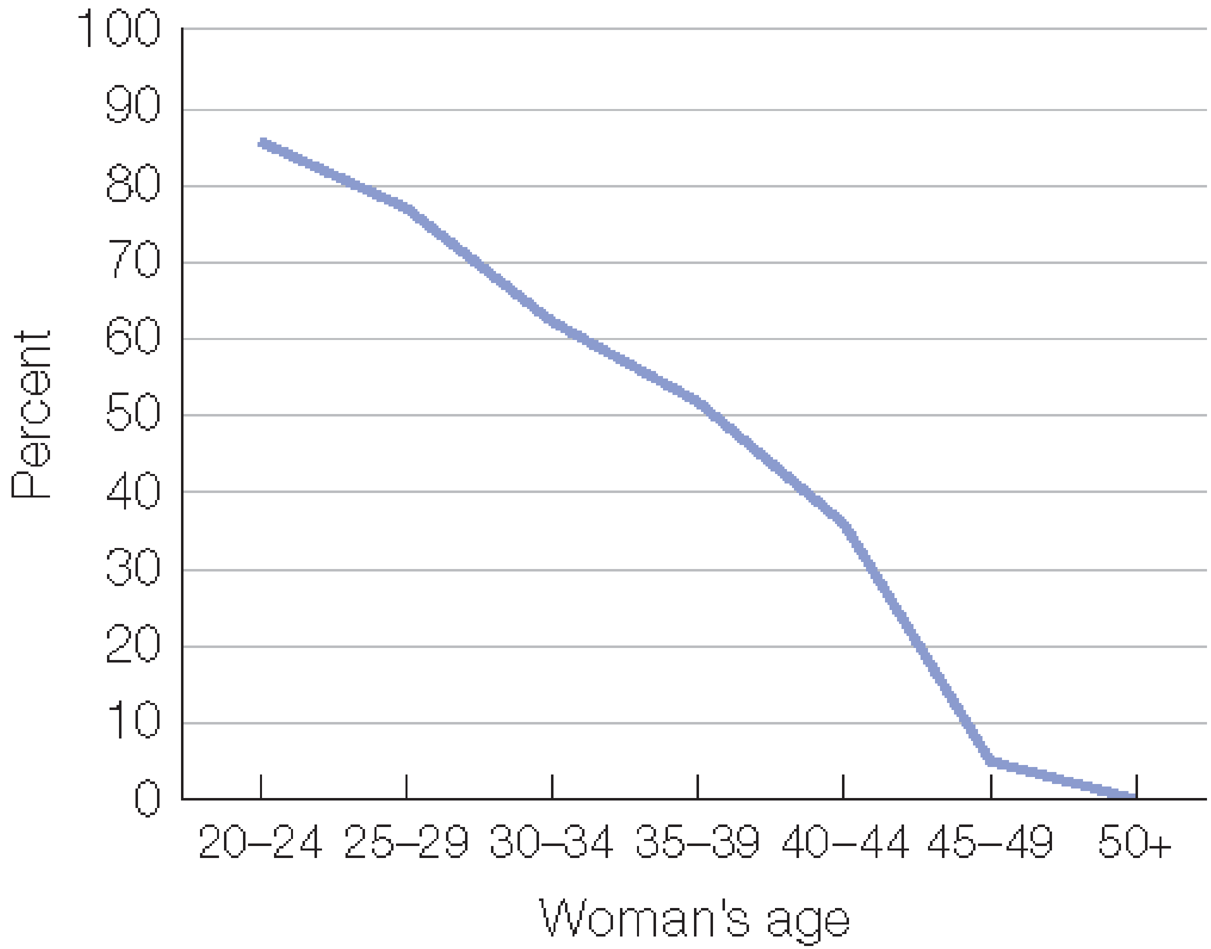


Figure 2.6 Fertility and Maternal Age



Why does fertility decline after the mid-20s?

Treatments for Infertility

- Many treatments utilize assisted reproductive technologies (ART)
  - Artificial insemination
    - Injects sperm into woman’s uterus
  - Fertility drugs
    - Mimic hormones involved in ovulation
  - In-vitro fertilization
    - Ova removed and fertilized outside the womb then placed into uterus

Stages of prenatal development

- Germinal – implantation in uterus
  - 0 to 2 weeks.
- Embryonic – placenta, umbilical cord, amniotic sac. Most critical periods-Organs are formed
  - 3-8 weeks.
- Fetal –

  Most sensitive periods- Organs mature
  - 9 weeks to birth.

  – From conception to birth: Visualized
  – https://youtu.be/fKyljukBE70?t=122

Stages of Prenatal Development

First Trimester				Second Trimester	Third Trimester
1–2 weeks	**3–4 weeks**	**5–8 weeks**	**9–12 weeks**	**13–24 weeks**	**25–38 weeks**
Germinal Zygote divides and forms blastocyst, which implants in uterus and begins forming the amnion, placenta, and umbilical cord	**Embryonic** Three layers form: the ectoderm, mesoderm, and endoderm; neural tube develops and produces neurons; heart begins beating; ribs, muscles, and digestive tract form	Arms and legs develop, then fingers and toes; placenta and umbilical cord function; digestive system develops; liver produces blood cells; embryo responds to touch	**Fetal** Genitals form and release sex hormones; fingernails, toenails, and taste buds develop; heartbeat audible with stethoscope	Mother feels movement; fetus kicks, turns, hiccups, sucks thumb, breathes amniotic fluid; responds to sounds, especially music and familiar voices; vernix and lanugo develop on skin	Lungs develop fully; over two-thirds of birth weight is gained; brain development accelerates; sleep-wake cycles resemble newborn's






Essentials of Prenatal care

- Diet: Balanced diet, lots of fluids. Gain between 25 and 35 pounds in total.

- Exercise: Mild to moderate exercise.

- Teratogens: Avoid tobacco, alcohol and other drugs.

Risk Factors During Prenatal Development

Major Risk Factors:

- Maternal and Environmental factors
- Interfere with normal prenatal development.
  - Death and birth defects (critical periods)
  - Abnormalities (sensitive periods).

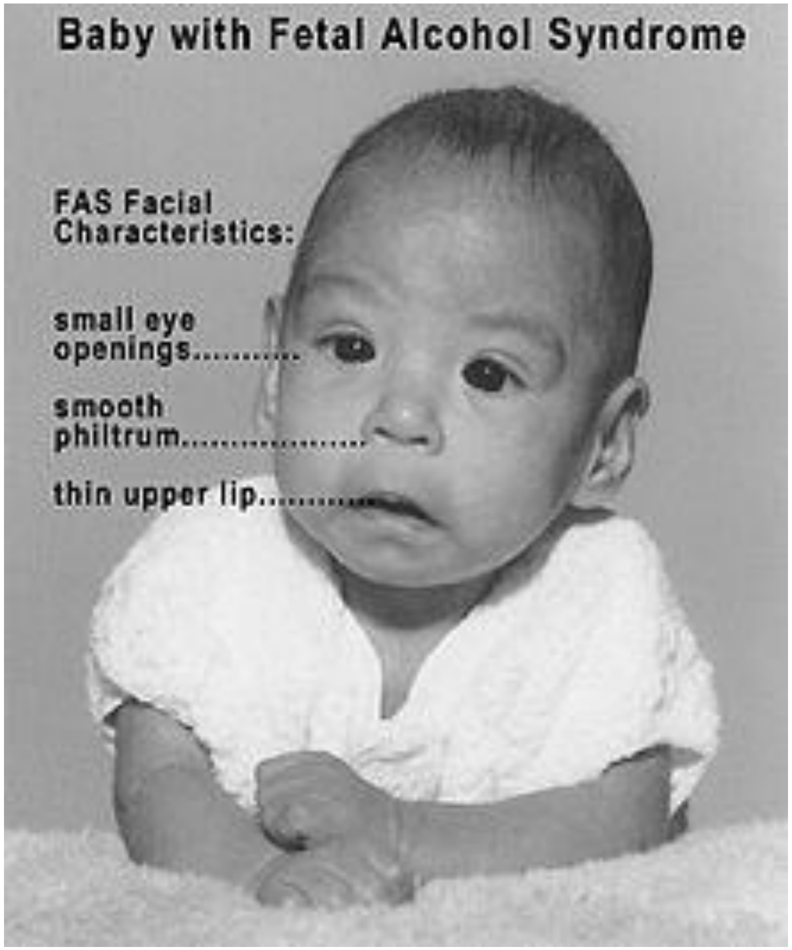
Can you name some common teratogens?

- Nicotine
- Alcohol
- Drugs
- Infectious diseases
- Malnutrition- Most common teratogen worldwide
- Stress

High-definition scans suggest effects of smoking may be seen in unborn babies

Movements in a fetus whose mother is a smoker (top) and a fetus whose mother is a non-smoker (below); credit Dr Nadja Reissland, Durham University

Fetal Alcohol Syndrome



- Intellectual impairment
- Facial deformities
- Slow growth rate

Infectious Diseases

- Rubella (German measles)
  - Exposure during embryonic stage can lead to heart abnormalities and intellectual disabilities.
  - Exposure during the fetal stage can lead to hearing problems and low birth weight
- Vaccination can help but rubella remains widespread in less developed countries

- AIDS (Acquired Immune Deficiency Syndrome)
- Three strategies can help prevent transmission
  - Effective medicines
  - Cesarean sections for AIDS-infected moms
  - Infant formula in place of breast-feeding

Experience in the Womb

WATCH: The impact of prenatal experiences

– *http://www.ted.com/talks/annie_murphy_paul_what_we_learn_before_we_re_born*

Test your learning: Can you answer these Qs?
(You do not have to turn these in but Qs may appear on test)

- According to Annie Murphy, when does learning begin?

- According to fetal origins research, are we clean slates when we are born?   Yes/No
- Researchers discovered that, after women repeatedly read aloud a section of Dr. Seuss' "The Cat in the Hat" while they were pregnant, their newborn babies____________ (recognized/did not recognize) that passage when they hear it outside the womb.

- How did researchers test whether fetuses recognize the taste of mother's food from before they were born?

- Name two important teratogens discussed in this video._____, ________

- Why did maternal malnourishment during pregnancy result in obesity later for children whose mothers were pregnant during the Hunger winter?

- What happened to the children whose mothers had experienced PTSD while they were in their mother's womb?

- What is the ultimate purpose of fetal learning? Does it serve any function for our survival?

Chromosomal Disorders

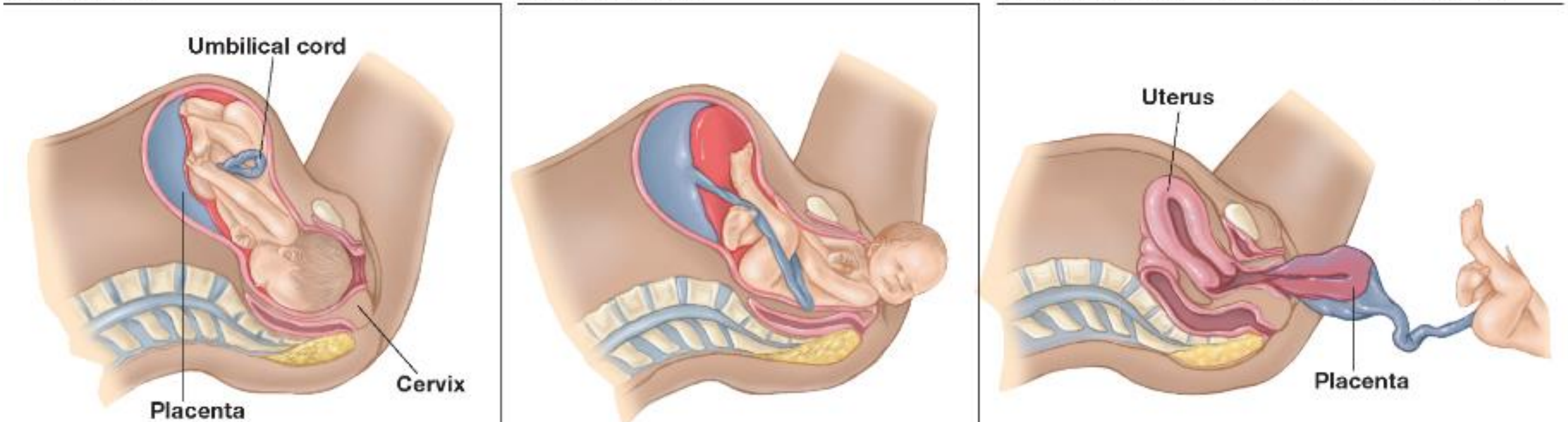
- Two types of chromosomal disorders are:
  - Sex chromosome disorders
  - Disorder on the 21^{st} chromosome (Down syndrome)

- Sex chromosomal disorder can result from an extra X, an extra Y, or only an X and no second chromosome
  - Common consequences of sex chromosome disorders include:
    - Cognitive deficit
    - Abnormality in reproductive system at puberty

Down Syndrome Trisomy-21

- Identifiable by physical characteristics
- Cognitive deficits
  - Speech problems
  - Mental retardation
- Social development varies
- Lower life expectancy

The Three Stages of the Birth Process



Stage 1: Labor

Contractions increase in duration, frequency, and intensity, causing the cervix to dilate.

Stage 2: Delivery

The mother pushes, and the baby crowns and then exits the birth canal and enters the world.

Stage 3: Expelling of Placenta & Umbilical Cord

Contractions continue as the placenta and umbilical cord are expelled.

Common complications

Two common birth complications:

- Failure to progress
- Breech presentation

Cesarean delivery can be done to deal with birth complications

Variations in the birthing methods

- As long as medical assistance is available in case of emergency, natural births and medical methods lead to same maternal and child outcomes (Bergstrom et al, 2009)
- Emotional and social support important
  - Father's presence linked to shorter labor and greater satisfaction for mother
- Medical use of epidural seen in developed countries
- Birthing position also eases pain
  - Upright, semi-sitting, half reclining position